Fermionic Open Strings as Pointlike Scatterers: Four-Ramond Amplitudes, Contact Limits, and a Quark Bootstrap

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(Dated: October 10, 2026)

We study a concrete operational question behind old hadronic bootstrap ideas: how would an extended spin-1/2 open string reveal itself in elastic scattering, and can it imitate a point fermion over a parametrically large kinematic domain? For a charged Ramond open-string state we derive a Møller-like four-fermion amplitude. In a minimal Chan–Paton embedding the complete disk result is the gauge-theory amplitude multiplied by a single Veneziano form factor, $\mathcal{M}_{RR} = V(t,u)\mathcal{M}_{\rm gauge}$, so the first oscillator correction is $O(\alpha'^2 E^4)$ rather than the $O(E^2/\Lambda^2)$ term characteristic of generic compositeness. We work out the exact angular distribution, the effective dimension-eight contact interaction, the crossed-channel Regge poles, and the high-energy fixed-angle suppression. We separately analyze the genuinely allowed exchange of a single fermionic mediator in mixed boson–fermion channels and its zero-range limit. We then revisit a quark–diquark bootstrap motivated by the antisymmetric equivalences $\bar{3}\otimes\bar{3}\supset 3$ in color and flavor, old hadronic supersymmetry, and modern light-front superconformal constructions. A scalar anti-diquark has precisely the color, light-flavor and electric charges of a quark; attaching a fermionic string supplies the required spin, while baryon number differs by one unit and must therefore reside in an additional string/junction topological sector. Recent baryon-junction evidence makes that assignment phenomenologically interesting. However, the common Regge slope of Neveu–Schwarz and Ramond sectors gives a sharp obstruction: a hadron-scale bosonic string cannot have a pointlike fermionic partner at multi-TeV momentum transfer if both are literally sectors of the same open string. The bootstrap can survive only as a dual/operator identification or through a mechanism separating the QCD flux-tube scale from the hard Ramond scattering scale. We formulate the resulting experimental tests and distinguish them from conventional compositeness searches.

I. INTRODUCTION

The first string models were not introduced as theories of quantum gravity but as analytic descriptions of hadronic scattering. Veneziano's amplitude [1], its N-point generalization and Chan–Paton endpoint factors [2], and the dual-resonance program embodied a particularly strong version of S-matrix bootstrap: poles seen as elementary exchanges in one channel were simultaneously generated as composite resonances in crossed channels. Ramond's fermionic dual model [3], the Neveu–Schwarz (NS) mesonic construction [4], and their later GSO projection [5, 6] showed that spacetime fermions and bosons can arise as different sectors of a single extended object.

A parallel hadronic line of thought anticipated part of the same algebra without fundamental superstrings. Miyazawa organized mesons and baryons into a graded multiplet using spinor and scalar triplets [7]. Aharonov, Casher and Susskind obtained the NS meson model and a fermionic baryon model from even and odd numbers of spin-1/2 partons and explicitly computed a meson–baryon amplitude [8]. Bars and Hanson constructed strings with spin-0 and spin-1/2 quark degrees of freedom at their ends and applied them to mesons and baryons [9]. Catto and Gürsey later showed how Miyazawa-like quark–diquark supersymmetry can emerge when the confining interaction becomes effectively stringlike [10]. Modern light-front superconformal QCD makes the relation especially explicit: the effective supercharge can be interpreted as replacing a quark by an anti-diquark in the same color representation [11, 12].

These observations motivate a sharper and experimentally meaningful question. Suppose that a fermion is represented, at least in one dual description, by two endpoint labels joined by a spinorial open string. Must elastic scattering expose its finite extension through a conventional form factor

$$F(q^2) = 1 - \frac{1}{6} r^2 q^2 + O(q^4), \tag{1}$$

so that compositeness appears at order E^2/Λ^2? Or can an extended fermionic string scatter as a point particle to higher order in energy?

The perturbative open superstring gives a precise answer. For massless states on the same brane stack, the four-point disk amplitude factorizes into a super-Yang–Mills (SYM) amplitude times an Euler/Veneziano function. At four points this universal factor begins at $O(\alpha'^2)$ [13–16]. Thus a string with characteristic scale $\sqrt{\alpha'}$ can be extended while its tree-level massless scattering shows no dimension-six oscillator correction. This behavior is known in D-brane phenomenology and contrasts with intersecting-brane matter, for which dimension-six four-fermion operators are generic [17].

The aim of this paper is fourfold. First, we derive the four-Ramond analogue of Møller scattering and turn the phrase "pointlike below the string scale" into an exact cross-section ratio. Second, we separate this elastic problem from the distinct question of whether a sin-

gle fermion can mediate an interaction, including the zero-range limit. Third, we connect the string result to quark–diquark/hadronic supersymmetry and formulate a concrete bootstrap map between a quark and a scalar anti-diquark plus a fermionic string. Fourth, we identify the obstruction that prevents a literal hadron-sized NS string from sharing a Ramond partner that remains pointlike at high momentum transfer.

Throughout we distinguish three levels of assertion:

(i) the four-Ramond disk amplitude and its low-energy expansion, which are standard perturbative string theory;

(ii) representation-theoretic quark–anti-diquark equivalences, which are standard QCD/hadronic-supersymmetry observations;

(iii) the identification of the observed quark with an anti-diquark plus a fermionic string, which is conjectural and subject to nontrivial dynamical constraints.

Keeping these levels separate is essential because the amplitude calculation is considerably stronger than a constituent analogy, whereas the proposed quark bootstrap is considerably stronger than established hadronic supersymmetry.

II. POINT-FERMION BENCHMARK: MASSLESS MØLLER SCATTERING

Consider

$$e^-(p_1)e^-(p_2) \to e^-(p_3)e^-(p_4), \tag{2}$$

with all external fermions taken massless. Define

$$s=(p_1+p_2)^2, \qquad t=(p_1-p_3)^2, \qquad u=(p_1-p_4)^2, \tag{3}$$

so that $s+t+u=0$. The tree amplitude is the antisymmetrized sum of t- and u-channel photon exchange,

$$\mathcal{M}_{\rm QED} = e^2\left(\frac{J_t}{t}-\frac{J_u}{u}\right), \tag{4}$$

where

$$\begin{aligned} J_t &= (\bar u_3\gamma^\mu u_1)(\bar u_4\gamma_\mu u_2),\\ J_u &= (\bar u_4\gamma^\mu u_1)(\bar u_3\gamma_\mu u_2). \end{aligned} \tag{5}$$

The relative minus sign is the exchange sign of identical final fermions.

Averaging over initial spins and summing over final spins gives

$$\overline{|\mathcal{M}_{\rm QED}|^2} = 2e^4\left[\frac{s^2+u^2}{t^2}+\frac{s^2+t^2}{u^2}+\frac{2s^2}{tu}\right]. \tag{6}$$

In the center-of-mass frame,

$$t=-\frac{s}{2}(1-\cos\theta), \qquad u=-\frac{s}{2}(1+\cos\theta), \tag{7}$$

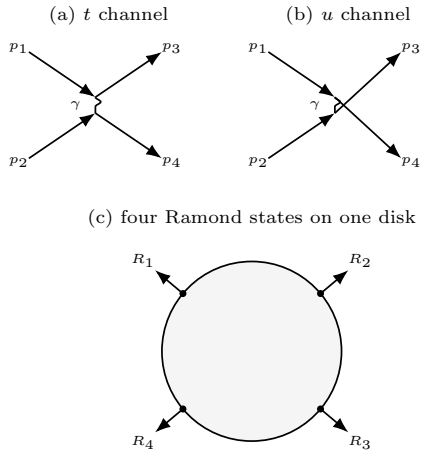


FIG. 1. Point-particle and string descriptions. Møller scattering is a sum of two Feynman graphs. The open-string tree amplitude is a single disk world sheet with four boundary Ramond insertions; its field-theory channels arise as degeneration limits in which neighboring insertions collide.

and inclusion of the 1/2! identical-particle factor yields

$$\boxed{\frac{d\sigma_{\rm QED}}{d\Omega} = \frac{\alpha^2}{2s}\frac{(3+\cos^2\theta)^2}{\sin^4\theta}.} \tag{8}$$

Appendix A records the trace reduction because the same current structure will reappear in the string result.

III. RAMOND OPEN STRINGS AND THE FOUR-FERMION DISK AMPLITUDE

A. World-sheet setup

For an open RNS string, a massless Ramond state in the $-1/2$ picture has vertex operator

$$V_R^{(-1/2)}(u,k;x) = g_o\,\lambda\,u^A S_A(x)\,e^{-\phi(x)/2}\,\Sigma(x)\,e^{ik\cdot X(x)}. \tag{9}$$

Here S_A is the spacetime spin field, Σ represents any internal compactification spin field, and λ is the Chan–Paton matrix. Four such vertices supply total superghost charge -2 on the disk. The ordered amplitude is

$$\mathcal{A}(1,2,3,4) = \int \frac{\prod_i dx_i}{\mathrm{vol}\,PSL(2,\mathbb{R})}\left\langle \prod_{i=1}^4 V_{R,i}^{(-1/2)}(x_i)\right\rangle. \tag{10}$$

Fixing $(x_1,x_3,x_4)=(0,1,\infty)$ leaves one modulus $x=x_2\in(0,1)$. The X^μ correlator contributes the Koba–Nielsen factor, while the spin-field and ghost correlators generate the spacetime fermion numerator.

A key simplification, transparent in the pure-spinor formulation and equivalent to the RNS result, is that the four-point massless open-superstring partial amplitude is

the SYM partial amplitude multiplied by a single Euler function [15, 16]:

$$\boxed{A_{\rm str}(1,2,3,4) = A_{\rm SYM}(1,2,3,4)\, V(s_{12}, s_{23}),} \tag{11}$$

where

$$V(a,b) = \frac{\Gamma(1-\alpha' a)\Gamma(1-\alpha' b)}{\Gamma[1-\alpha'(a+b)]}. \tag{12}$$

The statement is independent of whether the external states selected from the massless supermultiplet are bosons or fermions; supersymmetric Ward identities take the same form at disk level as in the field-theory limit [14].

The Gamma functions originate from an Euler integral. Suppressing a rational kinematic prefactor,

$$I(a,b) = \int_0^1 dx\, x^{-1-\alpha' a}(1-x)^{-1-\alpha' b} = B(-\alpha' a, -\alpha' b), \tag{13}$$

and

$$B(-\alpha' a, -\alpha' b) = \frac{\Gamma(-\alpha' a)\Gamma(-\alpha' b)}{\Gamma[-\alpha'(a+b)]}. \tag{14}$$

The spin-field numerator supplies the factors which convert this beta function into Eq. (11), with exactly the massless SYM poles at low energy. From the world-sheet viewpoint, the Feynman channels of Fig. 1(a,b) are not independent fundamental graphs: $x \to 0$ and $x \to 1$ are boundary components of the same disk moduli space.

B. Chan–Paton realization of a charged Ramond state

A single $U(1)$ gaugino is neutral, so it is not literally an electron. To isolate the kinematic comparison from model-building details, consider instead a coincident $U(2)$ stack and the off-diagonal Chan–Paton matrices

$$\lambda_+ = E_{12}, \qquad \lambda_- = E_{21}, \qquad Q = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{15}$$

They satisfy

$$[Q, \lambda_\pm] = \pm\lambda_\pm, \tag{16}$$

so the corresponding Ramond state is charged under the Cartan $U(1)$. This is a deliberately minimal toy embedding: it is an adjoint state of the unbroken non-Abelian theory, not a realistic Standard Model electron. Its virtue is that the Chan–Paton algebra cleanly projects the same-sign process onto the two exchange channels relevant for Møller scattering.

Cross the two outgoing positive-charge states to incoming negative-charge states. In an all-incoming convention take

$$\lambda_1 = \lambda_2 = E_{12}, \qquad \lambda_3 = \lambda_4 = E_{21}. \tag{17}$$

Because

$$E_{12}^2 = E_{21}^2 = 0, \tag{18}$$

any cyclic trace with equal charges adjacent vanishes, e.g.

$$\mathrm{Tr}(E_{12}E_{12}E_{21}E_{21}) = 0, \tag{19}$$

whereas alternating traces survive,

$$\mathrm{Tr}(E_{12}E_{21}E_{12}E_{21}) = 1. \tag{20}$$

Thus the only nonzero orderings are $(1,3,2,4)$ and $(1,4,2,3)$ (up to cyclicity and reflection). Their adjacent invariants are (t,u) and (u,t). Since $V(a,b)$ is symmetric, both carry the same string factor.

The full amplitude therefore takes the especially simple form

$$\boxed{\mathcal{M}_{RR} = V(t,u)\, \mathcal{M}_{\rm gauge},} \tag{21}$$

where $\mathcal{M}_{\rm gauge}$ has the same current-current structure as Eq. (4), modulo the chosen gauge-group normalization. If the Cartan coupling is denoted by e, then

$$\mathcal{M}_{RR} = e^2 V(t,u)\left(\frac{J_t}{t} - \frac{J_u}{u}\right). \tag{22}$$

This exact factorization is special to the alternating color assignment. A general non-Abelian four-fermion disk amplitude is a sum over orderings,

$$\mathcal{A}_4 = g_o^2 \sum_{\sigma/\mathbb{Z}_4} \mathrm{Tr}(\lambda_{\sigma_1}\lambda_{\sigma_2}\lambda_{\sigma_3}\lambda_{\sigma_4}) A_{\rm SYM}(\sigma) V_\sigma, \tag{23}$$

with different V_σ in different color channels. What remains universal ordering by ordering is the absence of an $O(\alpha')$ oscillator correction.

IV. EXACT OBSERVABLE AND LOW-ENERGY EXPANSION

In the physical same-sign region $s > 0$ and $t, u < 0$. Define

$$z \equiv \alpha' s = \frac{s}{M_s^2}, \qquad c \equiv \cos\theta, \tag{24}$$

with $M_s^2 \equiv 1/\alpha'$. Equations (7) and (12) give

$$\boxed{V(z,c) = \frac{\Gamma[1+\frac{z}{2}(1-c)]\,\Gamma[1+\frac{z}{2}(1+c)]}{\Gamma(1+z)}.} \tag{25}$$

Therefore

$$\boxed{\frac{d\sigma_{RR}}{d\Omega} = |V(z,c)|^2 \frac{d\sigma_{\rm gauge}}{d\Omega}.} \tag{26}$$

For the Abelian-normalized toy model, $d\sigma_{\rm gauge}/d\Omega$ is Eq. (8).

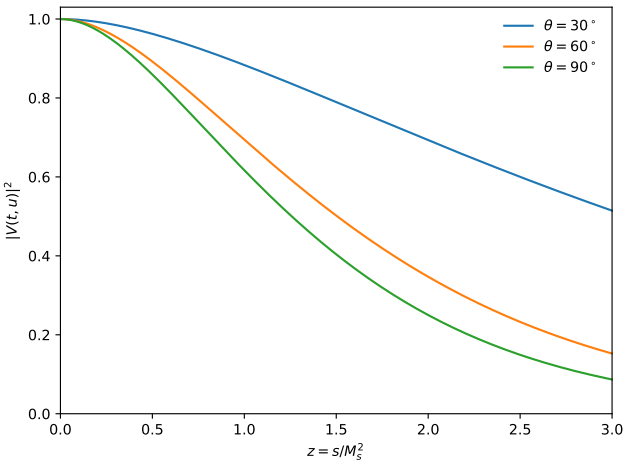


FIG. 2. Exact disk correction $|V(t,u)|^2$ to the Møller-like angular distribution as a function of $z = s/M_s^2$. The string is almost pointlike for $z \ll 1$, with the leading deviation proportional to $z^2 \sin^2\theta$. At fixed nonzero angle the suppression becomes strong as the string scale is approached.

The small-α' expansion follows from

$$\log\Gamma(1-x) = \gamma x + \sum_{n=2}^{\infty} \frac{\zeta(n)}{n} x^n. \tag{27}$$

For general a, b,

$$\log V(a,b) = \sum_{n=2}^{\infty} \frac{\zeta(n)}{n} \alpha'^n \left[a^n + b^n - (a+b)^n\right]. \tag{28}$$

Taking $(a,b) = (t,u)$ and using $t+u = -s$ gives

$$V(t,u) = 1 - \zeta(2)\alpha'^2 tu + \zeta(3)\alpha'^3 stu + O(\alpha'^4). \tag{29}$$

Hence

$$\mathcal{M}_{RR} = \mathcal{M}_{\rm gauge}\left[1 - \frac{\pi^2}{6}\frac{tu}{M_s^4} + \zeta(3)\frac{stu}{M_s^6} + \cdots\right], \tag{30}$$

and

$$\boxed{\frac{d\sigma_{RR}}{d\sigma_{\rm gauge}} = 1 - \frac{\pi^2}{12}\left(\frac{s}{M_s^2}\right)^2 \sin^2\theta + O\left(\frac{s^3}{M_s^6}\right).} \tag{31}$$

The first correction is quartic in E/M_s, not quadratic.

At $\theta = 90^\circ$ the exact expression reduces to

$$V_{90}(z) = \frac{\Gamma(1+z/2)^2}{\Gamma(1+z)}. \tag{32}$$

A few values are

z	0.10	0.25	0.50	1.0	2.0
$\lvert V_{90}\rvert^2$	0.992	0.957	0.859	0.617	0.250

so the transition from apparently pointlike to manifestly stringy is rapid once $\sqrt{s}$ approaches M_s.

A. Effective-field-theory interpretation

Multiplying Eq. (4) by the first correction in Eq. (29) gives

$$\delta\mathcal{M} = -\zeta(2)e^2\alpha'^2\,(uJ_t - tJ_u)\,. \tag{33}$$

The propagator poles have canceled. This is therefore a local contact term with two additional derivatives on each fermion bilinear in total. A convenient schematic operator basis is

$$\Delta\mathcal{L}_8 \sim \frac{g^2}{M_s^4}(\bar\psi\gamma^\mu D^\nu\psi)(\bar\psi\gamma_\mu D_\nu\psi) + \cdots, \tag{34}$$

with the omitted terms fixed by gauge invariance, Fierz identities and the supersymmetric completion. The important invariant statement is the dimension: the oscillator sector generates no tree-level $1/M_s^2$ four-fermion contact term in this same-stack supersymmetric amplitude.

This should not be confused with a theorem about all string compactifications. If chiral matter is localized at intersections of different D-brane stacks, exchange of winding modes, Kaluza–Klein excitations and string oscillators can produce dimension-six operators [17]. Nor does Eq. (31) say that a fundamental string has zero geometrical size. The target-space coordinate field X^μ has the usual oscillator fluctuations; indeed the light-cone mean-square size of an ideal fundamental string is resolution-dependent and logarithmically divergent [18]. "Pointlike through dimension six" is an on-shell statement about a particular scattering amplitude.

V. CROSSING, REGGE FACTORIZATION, AND HARD SCATTERING

Same-sign scattering has no physical string resonance in the simple Chan–Paton channel above. The surviving Veneziano factor depends on $t, u < 0$, and the s-channel ordering vanishes. Its experimental signature is therefore a smooth angular deformation rather than a resonance tower.

Crossed processes are different. When a color ordering contains a timelike invariant $a = s > 0$, Eq. (12) has poles at

$$\alpha' s = n, \qquad n = 1, 2, 3, \ldots. \tag{35}$$

Near the nth pole,

$$V(s,b) \sim \frac{(-1)^{n-1}}{(n-1)!}\frac{\prod_{k=0}^{n-1}(\alpha' b + k)}{\alpha'(s - nM_s^2)} \tag{36}$$

up to a convention-dependent overall sign. The degree-n residue polynomial is the standard signal that the exchanged level contains a tower of spins. This is the precise analytic continuation connecting subthreshold contact corrections with direct Regge production.

The opposite limit, $s \gg M_s^2$ at fixed angle, exposes the spatially extended nature of the string in a different way. Put

$$x = \frac{1-\cos\theta}{2}, \qquad 1-x = \frac{1+\cos\theta}{2}. \tag{37}$$

Then Eq. (25) is

$$V = \frac{\Gamma(1+zx)\Gamma[1+z(1-x)]}{\Gamma(1+z)}. \tag{38}$$

Stirling's formula yields

$$V \sim \sqrt{2\pi z x(1-x)}\,\exp[-zH(x)], \tag{39}$$

where

$$H(x) = -x\ln x - (1-x)\ln(1-x) > 0. \tag{40}$$

At 90°, $H = \ln 2$ and

$$V_{90} \sim \sqrt{\frac{\pi z}{2}}\,2^{-z}. \tag{41}$$

The resulting exponential fixed-angle softness is the open-string counterpart of the general high-energy behavior analyzed by Gross and Mende [19, 20]. Field-theory amplitudes fall by powers; strings eventually fall exponentially at fixed angle.

VI. CAN A SINGLE FERMION MEDIATE AN INTERACTION?

The four-Ramond result also clarifies a common categorical statement that "fermions cannot mediate forces." What is actually forbidden in a local relativistic action is a Grassmann-odd interaction term. A vertex with three fermion fields such as

$$\bar\psi\psi\chi \tag{42}$$

is not an ordinary bosonic term in the Lagrangian. Consequently a single fermion cannot generate a diagonal elastic potential $\psi\psi \to \psi\psi$ between unchanged fermionic sources through such a vertex.

But the Yukawa structure

$$\mathcal{L}_{\rm int} = g_1\phi_1^\dagger\bar\chi\psi_1 + g_2\phi_2^\dagger\bar\chi\psi_2 + \text{H.c.} \tag{43}$$

is perfectly Lorentz invariant and Grassmann even. A single fermion χ then mediates

$$\psi_1 + \phi_2 \to \phi_1 + \psi_2, \tag{44}$$

with

$$\mathcal{M}_\chi = g_1 g_2^*\,\bar u_2 \frac{\not q + M}{q^2 - M^2} u_1. \tag{45}$$

This mechanism is familiar in supersymmetric collider physics: for example, same-sign selectron production can proceed through t- and u-channel Majorana neutralino exchange [21, 22]. The external particle statistics change at each endpoint of the exchanged fermion line, while total fermion parity remains even.

The zero-range limit is especially transparent. For $|q^2| \ll M^2$,

$$\frac{\not q + M}{q^2 - M^2} = -\frac{1}{M} - \frac{\not q}{M^2} - \frac{q^2}{M^3} + \cdots, \tag{46}$$

so

$$\mathcal{L}_{\rm eff} = \frac{g_1 g_2^*}{M}\,\phi_1^\dagger\phi_2\,\bar\psi_2\psi_1 + \text{H.c.} + O(M^{-2}). \tag{47}$$

An exactly local representation is obtained with an auxiliary Grassmann field with no kinetic term,

$$\mathcal{L}_{\rm aux} = -M\bar\chi\chi + \bar\chi J + \bar J\chi, \tag{48}$$

where J is a fermionic composite current. Integrating out χ gives

$$\mathcal{L}_{\rm eff} = \frac{1}{M}\bar J J. \tag{49}$$

Thus zero range does not make fermionic mediation inconsistent; it turns the propagator into a local conversion operator.

The string version is the same selection rule geometrically. Elementary open-string three-point couplings are of NS–NS–NS and NS–R–R type. Therefore an $RR \to RR$ disk factorizes through NS states, while a mixed $R + NS \to NS + R$ process can factorize through a Ramond strip. The statement "forces are bosonic" is therefore too broad; the correct restriction concerns diagonal interactions between fixed-statistics external sectors.

VII. CONTACT RANGE DOES NOT ERASE INTERNAL EXCHANGE

A second misconception sometimes attached to the zero-range limit is that an S-wave contact interaction cannot distinguish a swap of endpoint labels. Only the *spatial* exchange factor becomes trivial at $L = 0$:

$$P_{\rm spatial} = (-1)^L \to +1. \tag{50}$$

The full exchange operator contains independent internal factors,

$$P = P_{\rm spatial}P_{\rm spin}P_{\rm flavor}P_{\rm color}\cdots. \tag{51}$$

For example, for two fundamental $SU(5)$ labels

$$5 \otimes 5 = 15_s \oplus 10_a, \tag{52}$$

so

$$P_{12}|15\rangle = +|15\rangle, \qquad P_{12}|10\rangle = -|10\rangle. \tag{53}$$

The strictly local interaction

$$V(\boldsymbol{r}) = g\,\delta^{(3)}(\boldsymbol{r})P_{12} \tag{54}$$

therefore produces opposite matrix elements in the two channels. More generally, contact interactions routinely contain

$$\delta^{(3)}(\boldsymbol{r})\,[c_0 + c_s\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 + c_f T_1^a T_2^a + c_P P_{\text{flavor}} + \cdots]\,. \tag{55}$$

Zero range removes spatial resolution, not internal representation dependence.

This point matters for the bootstrap interpretation below: even if the physical string scale is taken to infinity so that the extended object is replaced by a local spinor with two endpoint indices, the nontrivial content of those indices need not disappear. It survives as a representation-theoretic and contact-interaction structure.

VIII. ENDPOINT LABELS AND THE QUARK–ANTI-DIQUARK ALGEBRA

A. Color, flavor, charge, and spin

For QCD color,

$$3_c \otimes 3_c = \bar{3}_c \oplus 6_c, \qquad \bar{3}_c \otimes \bar{3}_c = 3_c \oplus \bar{6}_c. \tag{56}$$

Thus an antisymmetric anti-diquark can transform as a 3_c, exactly like a quark. The same statement holds for three light flavors,

$$\bar{3}_f \otimes \bar{3}_f = 3_f \oplus \bar{6}_f. \tag{57}$$

The fully antisymmetric light anti-diquarks may therefore be mapped as

$$[\bar{d}\bar{s}] \leftrightarrow u, \qquad [\bar{s}\bar{u}] \leftrightarrow d, \qquad [\bar{u}\bar{d}] \leftrightarrow s. \tag{58}$$

Their electric charges agree exactly:

$$Q_{\bar{d}\bar{s}} = +\frac{2}{3}, \qquad Q_{\bar{s}\bar{u}} = Q_{\bar{u}\bar{d}} = -\frac{1}{3}. \tag{59}$$

For an S-wave pair the Pauli principle is also favorable. Taking color antisymmetric and flavor antisymmetric makes the color-flavor product symmetric, so the spin wave function must be antisymmetric. The resulting anti-diquark is a scalar. Consequently

$$(\bar{q}\bar{q})_{3_c,3_f,S=0} \otimes (\text{fermionic string})_{S=1/2} \tag{60}$$

can carry the same $SU(3)_c \times SU(3)_f \times U(1)_{\text{em}}$ quantum numbers and total spin as a quark.

This is stronger than the observation that a diquark can act as an effective color source in a baryon. It says that the entire non-Abelian and electromagnetic label content of a light quark can be reproduced by an antisymmetric pair of anti-quark labels. What does *not* match is baryon number:

$$B(q) = \frac{1}{3}, \qquad B(\bar{q}\bar{q}) = -\frac{2}{3}, \tag{61}$$

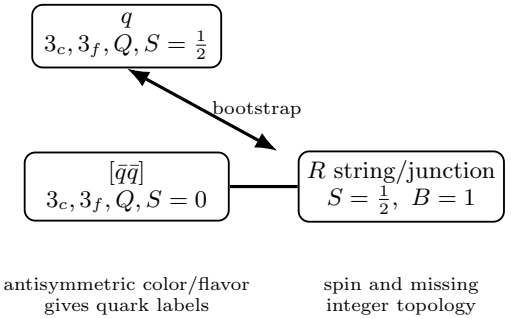


FIG. 3. Conjectural bootstrap map. The anti-diquark reproduces the quark's color, light-flavor and electric charge labels but is a scalar and has baryon number $-2/3$. A fermionic string/junction sector would have to supply spin $1/2$ and the missing unit of baryon number.

so

$$\Delta B = 1. \tag{62}$$

Any literal identification therefore requires an additional topological/string sector carrying one unit of baryon number.

B. Relation to hadronic supersymmetry

The algebraic core of Eq. (56) is precisely what underlies several historical and modern forms of hadronic supersymmetry. Miyazawa's construction connected spinor and scalar triplets [7]; Catto and Gürsey derived an effective quark–diquark supersymmetry from a stringlike confining interaction [10]; and light-front superconformal QCD relates meson and baryon wave functions through supercharges. In the latter language, the supercharge can be interpreted as transforming a quark into an anti-diquark in the same 3_c representation and an antiquark into a diquark in the $\bar{3}_c$ representation [12]. The construction also predicts baryon–meson–tetraquark supermultiplets, with the pion exceptional in the chiral limit rather than simply paired to a baryon [11].

The conjecture considered here goes one step further. Instead of treating quark and anti-diquark as different constituents connected by an effective supercharge, posit a bootstrap equivalence

$$\boxed{q \;\longleftrightarrow\; [\bar{q}\bar{q}]_{3_c,3_f,S=0} \otimes R_{B=1}.} \tag{63}$$

The Ramond-like string state $R_{B=1}$ supplies half-integer spin/statistics and the missing integer baryon topology. The claim is not that an ordinary spatially resolved bound state happens accidentally to mimic a point quark. The stronger version is that the two descriptions correspond to the same asymptotic state or operator sector, in the spirit of an S-matrix bootstrap or duality.

C. A useful $SU(5)$ endpoint bookkeeping example

Open strings naturally carry two endpoint labels. In an oriented sector one encounters $N \otimes \bar{N}$, while strings related to orientifold images can furnish symmetric or antisymmetric parts of $N \otimes N$ [23]. For $SU(5)$,

$$5 \otimes \bar{5} = 1 \oplus 24, \qquad 5 \otimes 5 = 10 \oplus 15. \tag{64}$$

If the fundamental is decomposed under $SU(2) \times SU(3) \times U(1)_X$ as

$$5 = (2,1)_3 \oplus (1,3)_{-2}, \tag{65}$$

then

$$15 = (3,1)_6 \oplus (2,3)_1 \oplus (1,6)_{-4}, \tag{66}$$

$$24 = (1,1)_0 \oplus (3,1)_0 \oplus (1,8)_0 \oplus (2,3)_{-5} \oplus (2,\bar{3})_5. \tag{67}$$

These identities illustrate that the "mesonic" $N\bar{N}$ and "diquark" NN label algebras required by a bootstrap can coexist naturally in open-string endpoint bookkeeping. They should not by themselves be read as a Georgi–Glashow grand-unified matter assignment; the point here is representation content, not a complete compactification.

IX. BARYON NUMBER AS A STRING/JUNCTION QUANTUM NUMBER

Equation (62) is the sharpest additive-quantum-number obstruction to Eq. (63). It also points toward a known alternative to assigning baryon number independently to each valence quark. In dual and large-N_c descriptions, baryons are associated with a Y-shaped color junction; Rossi and Veneziano developed this picture as part of a dual description of baryon dynamics [24]. The junction can transport baryon number over large rapidity intervals even when the valence quarks do not.

This issue has become experimentally timely. The STAR Collaboration recently measured observables in isobar and photonuclear collisions that are less naturally described if baryon number simply follows the valence quarks; the results favor significant baryon-number transport through a gluonic junction-like mechanism [25]. The data do not prove that a fundamental open string carries $B = 1$, nor do they establish the bootstrap in Eq. (63). They do, however, remove some of the conceptual artificiality from placing an integer baryonic quantum number in the confining-field topology rather than solely in endpoint labels.

There is an important geometrical caveat. The standard baryon junction is three-pronged, while Eq. (63) was motivated by a two-endpoint open string. A literal realization would therefore require either a hidden junction attached to the Ramond segment, a boundary-changing operator, or some other topological completion. This is a dynamical requirement, not a matter of charge bookkeeping.

X. QCD FLUX TUBES ARE NOT AUTOMATICALLY RNS STRINGS

The original dual models encouraged the identification of hadronic Regge trajectories with fundamental-like strings, but modern QCD effective string theory is more conservative. At long distance, a confining flux tube is described by the Goldstone modes of broken transverse translations, with universal terms in the long-string expansion [26]. Lattice spectra in pure gauge theory are broadly consistent with a bosonic effective string at sufficiently large length; classic studies already disfavored simple fermionic or NS universality classes for the ordinary QCD flux tube [27]. In supersymmetric gauge theories, world-sheet fermionic zero modes can exist, but they may be lifted as one deforms toward nonsupersymmetric QCD [28].

Thus there is no established statement that the QCD flux tube has a Ramond sector paired with the physical meson spectrum. What exists is a family resemblance between three structures:

(a) bosonic and fermionic sectors of perturbative RNS open strings;

(b) meson–baryon relations in hadronic supersymmetry;

(c) quark–diquark color equivalence in QCD.

Equation (63) proposes to identify them more strongly than present QCD derivations justify.

That stronger identification faces an immediate scale problem. NS and Ramond sectors of the same perturbative open string share the same target-space oscillators X^μ and the same tension

$$T = \frac{1}{2\pi\alpha'}. \tag{68}$$

A hadronic Regge slope $\alpha'_{\rm had} \simeq 0.9\ \mathrm{GeV}^{-2}$ corresponds to a string scale

$$M_s \sim (\alpha'_{\rm had})^{-1/2} \simeq 1.05\ \mathrm{GeV}. \tag{69}$$

If the very same α' controlled hard quark scattering through Eq. (26), the theory would fail spectacularly. At $\sqrt{s} = 10$ GeV and 90°, $z \simeq 90$, and Eq. (41) implies an amplitude suppression approximately 2^{-90} and a cross-section suppression approximately 2^{-180} apart from powers of z. Real quarks instead exhibit pointlike hard scattering at vastly larger momentum transfers.

This conclusion is stronger than a generic "electron and pion have different radii" argument. It uses the exact hard-scattering analytic form forced by a common open-string slope. A literal pion-scale NS string and a literal quark Ramond partner cannot simultaneously describe observed QCD hard scattering.

XI. EXPERIMENTAL MEANING

A. What should be measured?

For the simple Møller-like channel, the observable is the ratio

$$R(s,\theta) = \frac{(d\sigma/d\Omega)_{\rm string}}{(d\sigma/d\Omega)_{\rm point}} = |V(z,\cos\theta)|^2. \tag{70}$$

At low energy,

$$R - 1 \simeq -\frac{\pi^2}{12}\left(\frac{E_{\rm cm}}{M_s}\right)^4 \sin^2\theta. \tag{71}$$

The correction is largest in central scattering and vanishes in the forward limit. If a central-angle measurement agrees with pointlike theory to fractional precision ϵ, the parametric sensitivity is

$$M_s \gtrsim E_{\rm cm}\left(\frac{\pi^2}{12\epsilon}\right)^{1/4}. \tag{72}$$

The fourth root makes energy much more valuable than precision. At 1% one probes $M_s \simeq 3.0E_{\rm cm}$; at 0.1%, $M_s \simeq 5.35E_{\rm cm}$.

This signal is qualitatively distinct from the standard contact-interaction ansatz

$$\Delta\mathcal{L}_6 = \frac{\eta}{\Lambda^2}(\bar\psi\gamma_\mu\psi)(\bar\psi\gamma^\mu\psi), \tag{73}$$

which interferes with gauge exchange at relative order s/Λ^2. A quoted limit on an "electron radius" or quark compositeness scale derived from Eq. (1) or (73) cannot be converted into a limit on M_s by simply setting $\Lambda = M_s$.

LEP Bhabha analyses provide a useful historical example. A combined study of high-energy Bhabha data obtained a conventional electron-size bound $r_e < 2.8\times10^{-19}$ m and, in a specific TeV-string/large-extra-dimension model, a string-scale bound around 0.63 TeV [29]. Those two limits arose from different functional hypotheses and illustrate why "size" is model dependent.

B. Quark scattering and current collider constraints

For a quark realization the simplest robust expectation is not necessarily a single scalar factor multiplying the full QCD amplitude, because different color orderings in Eq. (23) can carry different Veneziano functions. Nevertheless, in a same-stack supersymmetric sector every ordering begins at $O(\alpha'^2)$, so one expects dimension-eight distortions of high-mass angular distributions before the first direct resonance.

Modern dijet measurements are already designed to test precisely such departures from Rutherford-like QCD angular behavior. The Particle Data Group summarizes dimension-six quark-compositeness limits at scales of tens of TeV [32]. A 2026 CMS analysis using the full Run-2 data set reports, for a benchmark left-handed four-quark operator, 95% confidence limits of roughly 17 TeV for destructive and 37 TeV for constructive interference [33]. These are impressive but they constrain Eq. (73), not Eq. (34). A dedicated fit to the energy and angular dependence of the string amplitude would be required.

Direct resonance searches constrain a complementary regime. Weakly coupled D-brane constructions predict universal first Regge excitations in partonic channels, motivating LHC dijet searches [30]. CMS excluded the benchmark "string resonance" model below about 7.7 TeV in a 13-TeV data set [31]. This is not a model-independent lower bound on the M_s appearing in every possible bootstrap construction; it assumes a particular weakly coupled production and branching pattern. It does show that a literal TeV-scale colored open string with conventional Regge couplings is already under severe pressure.

Experimentally the crossed-channel logic is therefore:

$$\begin{aligned}
\sqrt{\hat s} \ll M_s &: \quad \text{dimension-eight distortion},\\
\sqrt{\hat s} \sim M_s &: \quad \text{allowed-channel Regge poles},\\
\sqrt{\hat s} \gg M_s &: \quad \text{fixed-angle string softness}.
\end{aligned} \tag{74}$$

All three are analytic continuations of the same disk amplitude.

C. Discriminating string extension from ordinary compositeness

The experimentally relevant distinction is summarized in Table I. A conventional composite fermion is normally parameterized either by an elastic radius or by a dimension-six four-fermion operator. The same-stack open-superstring oscillator effect instead has no dimension-six term and predicts a correlated energy–angle dependence beginning at dimension eight. Intersecting-brane matter occupies a third category: although it is stringy microscopically, its low-energy phenomenology may look like ordinary dimension-six compositeness because winding and Kaluza–Klein exchange need not share the same cancellation [17].

For an e^-e^- machine the test would be unusually clean because Eq. (70) can be applied directly after standard electroweak and radiative corrections. As an illustration, at $\sqrt{s} = 3$ TeV and $\theta = 90°$, a string scale $M_s = 10$ TeV gives $z = 0.09$ and therefore a leading deficit of about

$$1 - R \simeq \frac{\pi^2}{12}z^2 \simeq 6.7\times10^{-3}. \tag{75}$$

The same fractional accuracy at 1 TeV would probe a scale three times smaller, illustrating directly the linear gain with collision energy in Eq. (72). Polarization would add another handle: because the string factor multiplies a definite gauge-theory helicity partial amplitude, ratios of polarized angular distributions can help distinguish

TABLE I. Schematic experimental diagnostics. These are leading tree-level scalings, not substitutes for a complete collider simulation.

Scenario	Leading relative effect	EFT order	Characteristic signal
ordinary radius	q^2r^2	dimension-6-like	form-factor distortion
contact compositeness	s/Λ^2	dimension 6	central excess or deficit
same-stack open string	$s^2 f(\theta)/M_s^4$	dimension 8	central suppression; crossed Regge poles
intersecting-brane string	often s/M_s^2	dimension 6 possible	contact terms plus string/KK states
hadronic-slope common RNS string	$\lvert V\rvert^2 \ll 1$ above GeV scale	nonlocal	excluded by hard parton scattering

a universal oscillator factor from a new chiral current-current operator.

At a proton collider the clean formula is folded with parton luminosities and with several QCD color orderings, so the optimal quantity is a normalized high-mass dijet angular distribution rather than an inclusive rate. For orientation only, the toy central-scattering factor at partonic $\sqrt{\hat{s}} = 6$ TeV would give a 2.1% leading deficit for $M_s = 15$ TeV and a 0.67% deficit for $M_s = 20$ TeV. These numbers are not direct LHC limits: PDFs, QCD higher orders, electroweak corrections, experimental migration and the non-Abelian color sum must all be included. They show, however, that a dimension-eight string signal need not be negligibly small in the highest-mass bins even when M_s lies several times above the partonic energy.

A dedicated analysis would therefore fit two logically linked structures rather than a generic contact coefficient. Below threshold it would use the exact or expanded Veneziano angular dependence; in channels with allowed timelike orderings it would simultaneously search for the first Regge pole near $\hat{s} = M_s^2$. Failure of these two inferred scales to agree would falsify the simplest weakly coupled open-string interpretation. Conversely, a smooth dimension-eight distortion with no correlated Regge structure would point toward a generic EFT operator rather than the specific analytic completion discussed here.

XII. WHAT WOULD COUNT AS A SUCCESSFUL BOOTSTRAP?

The preceding sections suggest a hierarchy of increasingly strong claims.

a. Level 1: representation equivalence. A scalar anti-diquark can have the same color, light-flavor and electric charge quantum numbers as a quark. This is exact group theory and already underlies quark–diquark phenomenology.

b. Level 2: dynamical hadronic supersymmetry. Meson and baryon spectra/wave functions can be related by an effective graded algebra in which a supercharge maps quarks to anti-diquarks. This is realized approximately in several hadronic models [10–12].

c. Level 3: fermionic string completion. A world-sheet Ramond degree of freedom attached to the anti-diquark endpoint pair supplies half-integer spin and statistics. Perturbative string theory demonstrates that such an extended fermionic object can scatter pointlike through dimension six.

d. Level 4: true bootstrap identity. The observed quark and the anti-diquark–string configuration are not merely superpartners or states with matching quantum numbers but two descriptions of the same asymptotic degree of freedom. At this level one must explain why no independent compositeness pole or form-factor radius appears, why baryon number is carried by the string/junction sector, and why the hadronic confinement scale does not force catastrophic high-energy softness of quark scattering.

The common-slope obstruction implies that a naive fundamental RNS implementation fails at Level 4. A viable theory would require at least one of the following:

(i) a genuinely dual description in which the hadronic flux tube and the hard Ramond object are not governed by the same α' even though they encode the same S-matrix state;

(ii) strong warping, compositeness, or renormalization of the effective hard scale that invalidates the naive common-tension disk amplitude;

(iii) a noncritical/effective world-sheet theory in which the hadronic bosonic string is not the NS sector of the fermionic bootstrap string;

(iv) a topological construction in which only endpoint/junction quantum numbers are supersymmetrically related, while the long QCD flux tube itself remains an ordinary bosonic effective string.

The last option is closest to established hadronic supersymmetry and least constrained by the RNS common-tension argument.

XIII. DISCUSSION

Two conclusions should be kept conceptually separate. The first is positive and rigorous within perturbative open-string theory: extension does not imply a leading radius-like correction to elastic scattering. A massless Ramond open string can have an exact tree amplitude whose first oscillator correction is dimension eight. In the particular charged $U(2)$ channel of Sec. III B, the statement is stronger still: the complete Møller-like amplitude

is the point-gauge amplitude multiplied by one universal function $V(t,u)$. The object is spatially extended, yet an experiment at $E \ll M_s$ would infer pointlike fermion behavior to order E^2/M_s^2.

The second conclusion is negative but equally sharp: this mechanism does not rescue a literal identification between a hadron-sized NS string and a high-energy pointlike quark if both share the same open-string slope. The same analytic form that delays the first low-energy correction also guarantees violent fixed-angle suppression once $E \gtrsim M_s$. With a hadronic value of M_s, QCD hard scattering would have disappeared at energies of only a few GeV.

This tension suggests a useful reinterpretation of old bootstrap language. The promising part is not "a quark is an ordinary extended bound state of two anti-quarks." A conventional bound state would generically possess a radius form factor, excited constituents, and additional poles. The stronger and more string-like possibility is an operator or S-matrix duality in which a quark channel can be reorganized into an anti-diquark plus topological flux configuration without introducing a new low-energy particle. Perturbative open-string amplitudes demonstrate that such reorganization can coexist with unusually pointlike low-energy scattering, while QCD phenomenology demands that the hard analytic scale be far above the hadronic Regge scale.

The baryon-number issue may ultimately be more informative than the spin issue. Spin 1/2 is easily supplied by a Ramond sector. In contrast, Eq. (62) forces the string/junction object to carry an integer topological charge. Baryon-junction phenomenology and the recent STAR measurements show that assigning baryon-number transport to gluonic topology is not merely a formal curiosity [24, 25]. A concrete bootstrap model would need to specify how that junction topology attaches to a two-endpoint fermionic object and how it behaves under string splitting/joining.

There is also a useful lesson for the original question of fermionic force carriers. The open-string world sheet makes the answer almost diagrammatically obvious. Four external fermionic strings factorize through a bosonic NS strip, exactly like two electrons exchanging a photon. But a mixed boson–fermion process can factorize through a Ramond strip, just as a Yukawa theory allows a fermion to be exchanged between vertices that convert bosons into fermions. Zero range preserves this distinction: the propagating fermion becomes a local off-diagonal contact operator. There is no principle forbidding fermionic mediation per se; there is a fermion-parity selection rule on the allowed external channels.

XIV. CONCLUSIONS

We have given an explicit scattering realization of a simple statement that is easy to obscure in qualitative discussions: an extended fermionic string can be experimentally more pointlike than a generic composite fermion. In a charged four-Ramond open-string channel,

$$\mathcal{M}_{RR} = V(t,u)\mathcal{M}_{\rm gauge}, \tag{76}$$

with

$$V(t,u) = \frac{\Gamma(1-\alpha' t)\Gamma(1-\alpha' u)}{\Gamma(1+\alpha' s)}. \tag{77}$$

The leading deviation from the point-fermion angular distribution is

$$\frac{\delta(d\sigma/d\Omega)}{d\sigma/d\Omega} = -\frac{\pi^2}{12}\frac{s^2}{M_s^4}\sin^2\theta + \cdots, \tag{78}$$

so no dimension-six oscillator contact term is present in the same-stack supersymmetric sector. Crossing exposes the associated Regge tower, and fixed-angle high-energy scattering exposes the expected exponential softness.

We have also shown that single-fermion exchange is allowed in mixed boson–fermion channels and has a well-defined local zero-range limit. What fails is only the use of a single fermion to generate a diagonal static force between unchanged ordinary fermionic sources.

Finally, we formulated a quark bootstrap that combines three exact or well-established ingredients—anti-diquark/quark representation equivalence, fermionic string states, and hadronic quark–diquark supersymmetry—with one conjectural identification. The anti-diquark can reproduce the quark's color, light-flavor, electric charge, and, once dressed by a fermionic string, spin. The missing baryon number is exactly one and points toward a junction/topological sector. The decisive constraint is scale separation: a literal common-tension NS/R string at the hadronic Regge slope is incompatible with observed hard quark scattering. Any viable realization must therefore be a dual/operator bootstrap or possess a mechanism that decouples the hard Ramond scattering scale from the ordinary QCD flux-tube scale.

Appendix A: Spin sums for the Møller benchmark

For completeness, write

$$\mathcal{M}_t = e^2\frac{J_t}{t}, \qquad \mathcal{M}_u = -e^2\frac{J_u}{u}. \tag{A1}$$

Using $\sum_s u(p,s)\bar{u}(p,s) = \not{p}$ and

$$\mathrm{Tr}(\not{a}\gamma^\mu\ \not{b}\gamma^\nu) = 4(a^\mu b^\nu + a^\nu b^\mu - g^{\mu\nu}a\cdot b), \tag{A2}$$

we obtain

$$\frac{1}{4}\sum|\mathcal{M}_t|^2 = 2e^4\frac{s^2+u^2}{t^2}, \tag{A3}$$

$$\frac{1}{4}\sum|\mathcal{M}_u|^2 = 2e^4\frac{s^2+t^2}{u^2}. \tag{A4}$$

For the interference term one may use a six-gamma trace or Fierz rearrangement; the result is

$$\frac{1}{4}\sum 2\Re(\mathcal{M}_t\mathcal{M}_u^*) = 4e^4\frac{s^2}{tu}. \tag{A5}$$

Combining terms gives Eq. (6). With Eq. (7),

$$\overline{|\mathcal{M}|^2} = 4e^4\frac{(3+\cos^2\theta)^2}{\sin^4\theta}. \tag{A6}$$

The two-body massless phase space gives $(64\pi^2 s)^{-1}\overline{|\mathcal{M}|^2}$, and the identical-final-state factor 1/2 yields Eq. (8).

Appendix B: Low-energy expansion of the Veneziano factor

Starting from Eq. (28), the first terms are

$$\begin{aligned}\log V(a,b) =& -\zeta(2)\alpha'^2 ab - \zeta(3)\alpha'^3 ab(a+b)\\ &- \zeta(4)\alpha'^4 ab\left(a^2+\frac{3}{2}ab+b^2\right)+O(\alpha'^5).\end{aligned} \tag{B1}$$

Exponentiating gives

$$\begin{aligned}V(a,b) =& 1-\zeta(2)\alpha'^2 ab - \zeta(3)\alpha'^3 ab(a+b)\\ &+\frac{1}{2}\zeta(2)^2\alpha'^4 a^2b^2\\ &-\zeta(4)\alpha'^4 ab\left(a^2+\frac{3}{2}ab+b^2\right)+O(\alpha'^5).\end{aligned} \tag{B2}$$

For $(a,b)=(t,u)$, $a+b=-s$, giving Eq. (29). The absence of an $O(\alpha')$ term follows before any helicity or color specialization and is therefore the robust feature relevant to same-stack massless disk amplitudes.

Appendix C: String scale versus conventional radius scale

It is useful to contrast two low-energy parameterizations. A conventional elastic form factor gives

$$F(q^2) = 1-\frac{q^2r^2}{6}+O(q^4), \tag{C1}$$

so a tree amplitude containing two such vertices generically changes by $O(q^2r^2)$. A current-current compositeness operator gives

$$\delta\mathcal{M}_6 \sim \frac{s}{\Lambda^2}. \tag{C2}$$

The same-stack open-string oscillator correction instead gives

$$\delta\mathcal{M}_8 \sim \frac{s^2}{M_s^4}f(\theta). \tag{C3}$$

Therefore an experimental upper limit on r or lower limit on Λ is not equivalent to a limit on $1/M_s$ even though all three quantities have dimensions of length or inverse energy. The energy power and angular dependence are part of the physical hypothesis.

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